

The j -formula: a structural refinement of the boundary lemma for the staircase Bott–Samelson rank-zero theorem

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Abstract

The May 8 rank-zero theorem reduced $\Pi^{S_n}|_{V_\lambda} = 0$ for $\ell(\lambda) > \lceil n/2 \rceil$ to a *boundary lemma*: there exists $j \in \{3, \dots, n\}$ with $B_j(V_\lambda) \subseteq E_1^-$, where $B_j := R'_j R'_{j+1} \cdots R'_n$. At $n \leq 6$ the lemma was verified directly; for $n \geq 8$ even it remained open.

We sharpen the lemma into a precise conjecture — the *j -formula*: the smallest such j is exactly $j(\lambda) = \ell(\lambda) + 1$. We verify the j -formula computationally for all rank-zero non-sign $\lambda \vdash n$, $n \leq 8$ (15 cases), and prove it structurally for the family $\lambda = (2, 1^{n-2})$ (all $n \geq 4$). As a corollary, the rank-zero theorem is now proven unconditionally for λ in the family $(2, 1^{n-2})$ at every n — a small but rigorous extension of the $n \leq 7$ result.

We further conjecture and computationally verify (at $n \leq 8$) the *rank-corner formula*:

$$\text{rank } B_{\ell+1}|_{V_\lambda} = \#\{r < \ell : \lambda_r > \lambda_{r+1}\},$$

which counts the corners of λ that preserve length when removed. This refines the j -formula by pinning down the precise dimension of the surviving image.

We close by presenting the inductive strategy and identifying the exact obstruction: the inductive step from a boundary $\lambda \vdash n$ to $\mu = \lambda^- \vdash n-1$ (length $\ell-1$, at threshold) requires a stronger property than the j -formula provides at $n-1$.

1 Setup and conventions

We retain notations from 2026-05-08-rank-zero-theorem.tex. $H_q(S_n)$ has generators $T_1, \dots, T_{n-1}$, V_λ is the seminormal cell module with basis $\{v_T\}$ indexed by SYTs T of shape λ . T_i acts in the seminormal basis with eigenvalues $\{q, -1\}$:

- If $i, i+1$ in the same row of T : $T_i v_T = q v_T$.
- If $i, i+1$ in the same column: $T_i v_T = -v_T$.
- Otherwise: T_i acts as a 2×2 Heffsmit block on $\text{span}(v_T, v_{s_i T})$ with eigenvalues $\{q, -1\}$.

$E_i^\pm$ denotes the $\pm$ -eigenspace of T_i in V_λ .

The staircase Bott–Samelson element is

$$\Pi^{S_n} = R'_2 R'_3 \cdots R'_n, \quad R'_k := (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1).$$

We write $B_j := R'_j R'_{j+1} \cdots R'_n$ for the right-tail of the staircase product.

2 The j -formula

2.1 Statement

Definition 1 (Length-preserving corner). A *corner* of λ is a cell (r, λ_r) with $\lambda_r > \lambda_{r+1}$ (with $\lambda_{\ell+1} = 0$). A corner is *length-preserving* if removing it does not decrease $\ell(\lambda)$, equivalently $r \leq \ell - 1$ (so that the row of length $\lambda_\ell \geq 1$ remains). Write $\kappa(\lambda)$ for the number of length-preserving corners.

Remark 2. For λ in the rank-zero regime, $\lambda_\ell = 1$ (proved below), so removing the row- ℓ corner $(\ell, 1)$ deletes the length-1 row, decreasing length by 1. Length-preserving corners are precisely those in rows $1, \dots, \ell - 1$.

Lemma 3 ($\lambda_\ell = 1$ in the rank-zero regime). *If $\lambda \vdash n$ has $\ell(\lambda) > \lceil n/2 \rceil$, then $\lambda_\ell = 1$.*

Proof. If $\lambda_\ell \geq 2$ then $|\lambda| \geq 2\ell > 2\lceil n/2 \rceil \geq n$, contradicting $|\lambda| = n$. $\square$

Conjecture 4 (j -formula). *Let $\lambda \vdash n$ with $\ell(\lambda) = \ell > \lceil n/2 \rceil$ and $\lambda \neq (1^n)$. Then*

$$B_{\ell+1}(V_\lambda) \subseteq E_1^-. \quad (1)$$

Moreover, $\ell + 1$ is sharp: $B_{\ell+2}(V_\lambda) \not\subseteq E_1^-$ in general (and is not contained in E_1^- in every case tested at $n \leq 8$).

Conjecture 5 (Rank-corner formula). *Under the hypotheses of Conjecture ??,*

$$\text{rank } B_{\ell+1}|_{V_\lambda} = \kappa(\lambda), \quad (2)$$

the number of length-preserving corners of λ .

The (1^n) sign-rep case is trivial: $\Pi^{S_n}|_{V_{(1^n)}} = 0$ since $T_i v = -v$ uniformly.

2.2 Computational verification at $n \leq 8$

Theorem 6 (Verification at $n \leq 8$). *Conjectures ?? and ?? hold for all rank-zero non-sign-rep $\lambda \vdash n$ with $4 \leq n \leq 8$ (15 cases). Moreover, for every such case, $B_{\ell+2}(V_\lambda) \not\subseteq E_1^-$ (so $\ell + 1$ is sharp).*

Verification. Computed in the seminormal basis at $q = 7 \in \mathbb{Q}$ (generic for V_λ) using SymPy. Results:

n	λ	ℓ	$\dim V_\lambda$	$\text{rank } B_{\ell+1}$	$\kappa(\lambda)$	$B_{\ell+2} \subseteq E_1^-?$
4	(2, 1, 1)	3	3	1	1	—
5	(2, 1, 1, 1)	4	4	1	1	—
6	(3, 1, 1, 1)	4	10	1	1	—
6	(2, 2, 1, 1)	4	9	1	1	—
6	(2, 1, 1, 1, 1)	5	5	1	1	—
7	(3, 1, 1, 1, 1)	5	15	1	1	no
7	(2, 2, 1, 1, 1)	5	14	1	1	no
7	(2, 1, 1, 1, 1, 1)	6	6	1	1	—
8	(4, 1, 1, 1, 1)	5	35	1	1	no
8	(3, 2, 1, 1, 1)	5	64	2	2	no
8	(2, 2, 2, 1, 1)	5	28	1	1	no
8	(3, 1, 1, 1, 1, 1)	6	21	1	1	no
8	(2, 2, 1, 1, 1, 1)	6	20	1	1	no
8	(2, 1, 1, 1, 1, 1, 1)	7	7	1	1	—

(Entries marked “—” have $\ell + 2 > n$, so $B_{\ell+2}$ is the identity, which trivially does not lie in E_1^- unless $V_\lambda = E_1^-$, i.e., the sign rep.)

Scripts: `~/projects/scratch/2026-05-11-j-formula/verify_j_formula.py, test_rank_corner_count.py.`

□

2.3 Implications for the rank-zero theorem

Corollary 7. *Conjecture ?? implies the boundary lemma (2026-05-08-rank-zero-theorem.tex, Lemma 4) at every n even, hence the rank-zero theorem unconditionally for all n .*

Proof. $B_{\ell+1}(V_\lambda) \subseteq E_1^-$ implies $\Pi^{S_n} V_\lambda = R'_2 \cdots R'_\ell \cdot B_{\ell+1}(V_\lambda)$ where the rightmost factor of R'_ℓ is (T_1+1) , which annihilates E_1^- . □

So the j -formula is the precise tool needed to close the rank-zero theorem. It also gives, via the rank-corner formula, finer information: not only does Π^{S_n} vanish, but $B_{\ell+1}$ already collapses V_λ to a $\kappa(\lambda)$ -dimensional subspace of E_1^- .

3 Structural proof for the family $\lambda = (2, 1^{n-2})$

We now prove the j -formula structurally for the simplest non-trivial family: partitions with exactly one cell outside column 1.

Theorem 8 (j -formula for $\lambda = (2, 1^{n-2})$). *For every $n \geq 4$, $\lambda = (2, 1^{n-2}) \vdash n$ has $\ell = n - 1$ and $j = \ell + 1 = n$, so $B_j = R'_n$. We have*

$$R'_n(V_{(2, 1^{n-2})}) \subseteq E_1^-.$$

Moreover the image is one-dimensional, matching $\kappa(\lambda) = 1$.

3.1 The seminormal basis of $V_{(2, 1^{n-2})}$

For $\lambda = (2, 1^{n-2})$, the $n - 1$ SYTs are parameterized by the value $a + 1 \in \{2, 3, \dots, n\}$ placed at position $(1, 2)$: $T^{(a)}$ ($a \in \{1, \dots, n - 1\}$) has

- $(1, 1) = 1, (1, 2) = a + 1$;
- $(b, 1) = b$ for $2 \leq b \leq a$ (only if $a \geq 2$);
- $(b, 1) = b + 1$ for $a + 1 \leq b \leq n - 1$.

Equivalently, the values $\{2, \dots, n\} \setminus \{a + 1\}$ fill column 1 (cells $(2, 1), (3, 1), \dots, (n - 1, 1)$) in increasing order.

Write $v_a := v_{T^{(a)}}$ for short. Then $E_1^- = \text{span}(v_2, v_3, \dots, v_{n-1})$: $T_1 v_a = q v_a$ iff $a = 1$ (value 2 at $(1, 2)$, same row as 1); $T_1 v_a = -v_a$ for $a \geq 2$ (values 1, 2 both in column 1).

3.2 Action of T_k on $V_{(2, 1^{n-2})}$

Lemma 9 (Action of T_k). *For $1 \leq k \leq n - 1$ and $a \in \{1, \dots, n - 1\}$:*

- If $a \notin \{k - 1, k\}$, then $T_k v_a = -v_a$.
- If $a = k - 1$ (so $k \geq 2$), then $\{T^{(k-1)}, T^{(k)}\}$ form a T_k -block of axial distance $-k$.
- If $a = k$, then $\{T^{(k-1)}, T^{(k)}\}$ form a T_k -block of axial distance $+k$ (the same block, viewed from $T^{(k)}$).

- *Special case $k = 1$: only $a = 1$ is in the block range; T_1 is diagonal with $T_1 v_1 = q v_1$ and $T_1 v_a = -v_a$ for $a \geq 2$.*

Proof. Locate k and $k + 1$ in $T^{(a)}$ using the parameterization above.

Case $a < k - 1$ (equivalently $a + 1 \leq k - 1 < k < k + 1$). Both k and $k + 1$ are larger than the value $a + 1$ at $(1, 2)$, so by the parameterization k lies at $(k - 1, 1)$ and $k + 1$ at $(k, 1)$ (column 1, adjacent cells). Same column, so $T_k v_a = -v_a$.

Case $a > k$ (equivalently $a + 1 \geq k + 2 > k + 1 > k$). Both k and $k + 1$ are smaller than $a + 1$. By the parameterization, k lies at $(k, 1)$ and $k + 1$ at $(k + 1, 1)$ (column 1, adjacent). $T_k v_a = -v_a$.

Case $a = k - 1$ (so $a + 1 = k$). Value k is at $(1, 2)$ (content 1); value $k + 1$ is at $(k, 1)$ (content $1 - k$). Different positions; axial distance $d = (1 - k) - 1 = -k$. The s_k -image $T^{(a)} \rightarrow T^{(k)}$ swaps $k \leftrightarrow k + 1$, giving $k + 1$ at $(1, 2)$, k at $(k, 1)$ — the SYT $T^{(k)}$.

Case $a = k$ (so $a + 1 = k + 1$). Symmetric: k at $(k, 1)$ (content $1 - k$), $k + 1$ at $(1, 2)$ (content 1); $d = k$; $s_k T^{(k)} = T^{(k-1)}$.

Special case $k = 1$. “ $a = k - 1 = 0$ ” is excluded; only $a = 1$ matches $a = k$, but the would-be block partner $T^{(0)}$ does not exist (value 1 is fixed at $(1, 1)$). So T_1 is diagonal: $T_1 v_1 = q v_1$ (1, 2 same row in $T^{(1)}$), and $T_1 v_a = -v_a$ for $a \geq 2$ (1, 2 same column). $\square$

3.3 Tracking the image of R'_n

Proposition 10 (The shifting window). *Apply $R'_n = (T_{n-1}+1)(T_{n-2}+1) \cdots (T_1+1)$ to $V_{(2,1^{n-2})}$ factor by factor, right-to-left. Write W_k for the image after applying the first k factors $(T_k+1) \cdots (T_1+1)$. Then*

$$W_k = \text{span}(v_{T^{(k)}}^+) \quad \text{for } k \geq 1, \quad (3)$$

where $v_{T^{(k)}}^+ := \alpha_k v_{T^{(k-1)}} + \beta_k v_{T^{(k)}}$ is the (unique up to scalar) $+q$ -eigenvector of T_k inside the 2-block $\{T^{(k-1)}, T^{(k)}\}$, with the convention that for $k = 1$ we set $v_{T^{(1)}}^+ := v_{T^{(1)}}$ (the block is degenerate; the “virtual” $T^{(0)}$ does not contribute).

In particular $W_{n-1} = R'_n(V_{(2,1^{n-2})})$ is one-dimensional and contained in $\text{span}(v_{T^{(n-2)}}, v_{T^{(n-1)}}) \subseteq E_1^-$.

Proof. Induction on k .

Base case $k = 1$. By Lemma ??, T_1 is diagonal with $T_1 v_1 = q v_1$ and $T_1 v_a = -v_a$ for $a \geq 2$. Thus $(T_1+1)v_1 = (1+q)v_1$, $(T_1+1)v_a = 0$ for $a \geq 2$. Image $W_1 = \text{span}(v_1)$. (Setting $v_{T^{(1)}}^+ := v_1$ matches.)

Inductive step. Assume $W_{k-1} = \text{span}(v_{T^{(k-1)}}^+)$ where $v_{T^{(k-1)}}^+ = \alpha_{k-1} v_{T^{(k-2)}} + \beta_{k-1} v_{T^{(k-1)}}$ (or $= v_1$ if $k - 1 = 1$). We compute $(T_k+1)v_{T^{(k-1)}}^+$.

By Lemma ??, $T_k v_{T^{(k-2)}} = -v_{T^{(k-2)}}$ since $k - 2 \notin \{k - 1, k\}$. So $(T_k+1)v_{T^{(k-2)}} = 0$.

For $v_{T^{(k-1)}}$: $a = k - 1$ matches $a = k - 1$ in Lemma ??, so $\{T^{(k-1)}, T^{(k)}\}$ form a T_k -block. $(T_k+1)v_{T^{(k-1)}} \in E_k^+$ within this block, i.e., a multiple of $v_{T^{(k)}}^+ := \alpha_k v_{T^{(k-1)}} + \beta_k v_{T^{(k)}}$, the $+q$ -eigenvector of T_k .

Combining: $(T_k+1)v_{T^{(k-1)}}^+ = \alpha_{k-1} \cdot 0 + \beta_{k-1} \cdot c_k \cdot v_{T^{(k)}}^+$ for some nonzero scalar $c_k \in \mathbb{Q}(q)$. So $W_k = \text{span}(v_{T^{(k)}}^+)$, completing the induction.

Conclusion. $W_{n-1} = \text{span}(v_{T^{(n-1)}}^+) \subseteq \text{span}(v_{T^{(n-2)}}, v_{T^{(n-1)}})$. Both $T^{(n-2)}$ and $T^{(n-1)}$ have $a \geq 2$, hence $T_1 v_{T^{(a)}} = -v_{T^{(a)}}$. So $W_{n-1} \subseteq E_1^-$. $\square$

Proof of Theorem ??. Direct from Proposition ??: $R'_n(V_{(2,1^{n-2})}) = W_{n-1}$ is one-dimensional and contained in E_1^- . Sharpness of $j = n$ for this family is immediate: $j = n + 1$ corresponds to no factors, and $V_{(2,1^{n-2})} \not\subseteq E_1^-$ for $n \geq 3$ (the SYT $T^{(1)}$ has $T_1 v_1 = q v_1$). $\square$

Corollary 11. $\Pi^{S_n}|_{V_{(2,1^{n-2})}} = 0$ for every $n \geq 2$.

Proof. $n = 2$: $V_{(1,1)}$ is the sign rep, $\Pi^{S_2} = T_1 + 1 = 0$. $n \geq 3$: by Theorem ??, $R'_n(V_{(2,1^{n-2})}) \subseteq E_1^-$. The next factor of Π^{S_n} is the rightmost (T_1+1) of R'_{n-1} (which exists for $n \geq 3$), and $(T_1+1)|_{E_1^-} = 0$. Hence $\Pi^{S_n}V_{(2,1^{n-2})} = 0$. $\square$

This is a small but real strengthening of the May-8 results: the rank-zero theorem on the family $(2, 1^{n-2})$ now has a structural proof valid for every $n \geq 2$, not contingent on the boundary lemma at $n = 2k$.

4 Inductive strategy and the obstruction

The natural inductive strategy for the j -formula at general λ uses the splitting $\Pi^{S_n} = \Pi^{S_{n-1}} \cdot R'_n$ and the branching $V_\lambda \downarrow_{H_q(S_{n-1})} = \bigoplus_{\mu \prec \lambda} V_\mu$:

$$B_{\ell+1}^{(n)} v = B_{\ell+1}^{(n-1)}(R'_n v) = \sum_{\mu \prec \lambda} B_{\ell+1}^{(n-1)} w_\mu, \quad w_\mu \in V_\mu \subset V_\lambda.$$

The inductive hypothesis: for every rank-zero $\nu \vdash n-1$ with $\ell(\nu) \leq \lceil (n-1)/2 \rceil$ + whatever, (??) holds at level $n-1$ with $j = \ell(\nu) + 1$.

4.1 Where induction works (deep regime)

For $\mu \prec \lambda$ with $\ell(\mu) = \ell(\lambda) =: \ell$ (corner removed from a row $r < \ell$): the j -formula at $n-1$ requires $\ell > \lceil (n-1)/2 \rceil$, which holds whenever $\ell > \lceil n/2 \rceil$ (in both parities). IH directly gives $B_{\ell+1}^{(n-1)}V_\mu \subseteq E_1^-$.

4.2 The obstruction: removing the row- ℓ corner

For $\mu = \lambda^- = (\lambda_1, \dots, \lambda_{\ell-1})$ obtained by removing the row- ℓ corner $(\ell, 1)$ (recall $\lambda_\ell = 1$ in the rank-zero regime, by Lemma ??): $\ell(\mu) = \ell - 1$.

Deep regime ($\ell \geq \lceil n/2 \rceil + 2$). $\ell - 1 \geq \lceil n/2 \rceil + 1 \geq \lceil (n-1)/2 \rceil + 1$, so μ is in the rank-zero regime at $n-1$. But the j -formula at $n-1$ for μ provides $B_{\ell(\mu)+1}^{(n-1)}V_\mu = B_\ell^{(n-1)}V_\mu \subseteq E_1^-$. We need the *stronger* statement $B_{\ell+1}^{(n-1)}V_\mu \subseteq E_1^-$, which uses one fewer staircase block.

Boundary regime ($n = 2k$, $\ell = k + 1$). $\ell - 1 = k = \lceil (n-1)/2 \rceil$, so μ is at the rank-zero *threshold* at $n-1$ — not in rank-zero regime. V_μ has rank $\Pi^{S_{n-1}} \geq 1$. The j -formula at $n-1$ does not apply.

4.3 What stronger property is needed

Define

$$\tilde{B}_{\ell+1}^{(n-1)}V_\mu := R'_{\ell+1} \cdots R'_{n-1}V_\mu$$

(one fewer factor than $B_\ell^{(n-1)} = R'_\ell \tilde{B}_{\ell+1}^{(n-1)}$). The induction goes through if and only if

$$\tilde{B}_{\ell+1}^{(n-1)}W_\mu \subseteq E_1^-, \tag{4}$$

where $W_\mu = \text{proj}_{V_\mu}(R'_n V_\lambda) \subseteq V_\mu$ is the projection of the image of R'_n onto the V_μ -summand.

Computational evidence. At $n = 8$, $\lambda = (4, 1^4)$, $\mu = \lambda^- = (4, 1^3)$, $\dim V_\mu = 20$. The image W_μ has $\dim < 20$ in general (the projection of a specific subspace of V_λ). Without computing W_μ explicitly, the conjecture is that the projection lands in $\ker \tilde{B}_{\ell+1}^{(n-1)}$ on V_μ/E_1^- .

4.4 What this proof provides, what it does not

- **Provides.** A precise formulation of the j -formula (Conjecture ??) that sharpens the boundary lemma. Computational verification at $n \leq 8$ (Theorem ??). A structural proof for the $(2, 1^{n-2})$ family at every n (Theorem ??). An explicit conjectural rank formula (Conjecture ??).
- **Does not provide.** A proof of the j -formula at general λ , even for the $n = 8$ boundary cases $(4, 1^4), (3, 2, 1^3), (2, 2, 2, 1, 1)$. The obstruction is the failure of the inductive step at $\mu = \lambda^-$ (length $\ell - 1$, at threshold or just above). The inductive step at the row- r corners ($r < \ell$) goes through.
- **Note.** The j -formula is a strictly stronger claim than the rank-zero theorem. The rank-zero theorem follows once any j works; the j -formula identifies the smallest such j . A direct proof of the boundary lemma (*any* j exists) would suffice for rank-zero, even without identifying the optimal j .

5 Open questions raised

1. **Prove the j -formula for the family $\lambda = (k, 1^{n-k})$ (single non-column-1 row) at general n .** This generalizes Theorem ??. The SYTs are parameterized by $\binom{n-1}{k-1}$ subsets, and the shifting argument should extend, but the multi-step interaction is more involved.
2. **Prove the j -formula for $\lambda = (2^j, 1^{n-2j})$ (rectangle-plus-tail).** These are the simplest non-hook tall partitions. The first non-hook case in the rank-zero regime is at $n = 10$, $\lambda = (3, 3, 1^4)$, where Conjecture ?? predicts rank 1 while the number of SYTs of $\lambda^{>1} = (2, 2)$ is 2.
3. **Identify the structural reason for the rank-corner formula.** The image of $B_{\ell+1}|_{V_\lambda}$ has dimension equal to the number of corners in rows $1, \dots, \ell - 1$. Find an explicit basis of the image indexed by these corners.
4. **Strengthen (??) to bridge the boundary case.** The natural sub-conjecture is: for $\mu \vdash n-1$ at the rank-zero threshold and $W_\mu \subset V_\mu$ the projection of $R'_n V_\lambda$ to V_μ , $\tilde{B}_{\ell+1}^{(n-1)} W_\mu \subseteq E_1^-$.

6 Honest assessment

What this paper proves.

1. The j -formula and rank-corner formula at every rank-zero non-sign $\lambda \vdash n$ for $4 \leq n \leq 8$ (Theorem ??, computational).
2. The j -formula structurally for $\lambda = (2, 1^{n-2})$ at every $n \geq 4$ (Theorem ??).
3. Hence, $\Pi^{S_n}|_{V_{(2, 1^{n-2})}} = 0$ unconditionally for every $n \geq 2$ (Corollary ??).

What is conjectural. The j -formula at general λ rank-zero (and the rank-corner formula). The boundary cases $(4, 1^4), (3, 2, 1^3), (2, 2, 2, 1, 1)$ at $n = 8$ are verified computationally but not proved structurally.

The obstruction. The branching $V_\lambda \downarrow_{S_{n-1}}$ contains the summand V_{λ^-} where λ^- has length $\ell - 1$. In the boundary regime ($n = 2k$, $\ell = k + 1$) this λ^- is at the rank-zero threshold at $n - 1$, where the j -formula does not apply. In the deep regime ($\ell \geq \lceil n/2 \rceil + 2$) the inductive hypothesis at $n - 1$ for λ^- provides the wrong index ($j = \ell$, not $\ell + 1$); a strengthening of the IH is needed.

Why a single structural family. The $(2, 1^{n-2})$ proof exploits a 1-parameter shifting structure in the seminormal basis $(T^{(a)})$ indexed by the value at the unique non-column-1 cell). The same idea should extend to other “narrow” families, but the shifting becomes multi-dimensional once λ has ≥ 2 cells outside column 1, requiring a more delicate combinatorial argument.

Computational note. All matrix computations use SymPy in the seminormal basis at $q = 7 \in \mathbb{Q}$ (generic). The check “ $M \subseteq E_1^-$ ” is performed column-by-column: for each column c and each row r corresponding to an SYT T with $2 \in (1, 2)$, verify $M[r, c] = 0$. Scripts at `~/projects/scratch/2026-05-11-j-formula/`.